

A High-Performance WSe₂/h-BN Photodetector using a Triphenylphosphine (PPh₃)-Based n-Doping Technique

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Atomically thin transition metal dichalcogenides (TMDs) with a 2D semiconducting layered structure have recently come into the spotlight as suitable materials for next-generation electronics^[1–4] and optoelectronics.^[5–7] For electronic device applications, due to their excellent scalability to a monolayer based on its van der Waals epitaxial structure, TMDs have high immunity to a short channel effect, providing the possibility for achieving ultrascaled transistors with superior gate controllability.^[8] In addition, a smooth surface without dangling bonds and thereby native oxides results in high carrier mobilities above 100 cm² V^{−1} s^{−1}, even though the TMDs are sub-1 nm thick semiconductors.^[1,9] Alternatively, TMDs also exhibit strong optical absorption (i.e., quantum efficiency: 45% for MoS₂^[10] and 80% for WSe₂^[11]) and ultrafast charge transfer (i.e., 5 × 10^{−14} s for the hole transport from MoS₂ to WSe₂^[12]), providing many opportunities for photodetection and photovoltaics. However, since one type of transistor can normally be fabricated on TMDs due to the Fermi-level pinning phenomenon between metals and TMDs^[13,14] (i.e., n-channel for MoS₂ and p-channel for WSe₂), it is currently difficult to expand research regarding various TMD-based electronic and optoelectronic applications. In this light, recently reported works have focused on the fabrication of TMD devices that operate in the opposite way (p-channel in MoS₂ and n-channel in WSe₂) by reducing the high Schottky barriers at the metal-TMD contacts. Javey et al. reported the MoS₂ p-FET^[15] and WSe₂ n-FET^[16] by inserting a MoO₃ layer and applying a potassium doping technique at the metal-TMD junction, respectively. They also

demonstrated the inverter circuit operation with complementary FETs fabricated on the same WSe₂ flake, where different contacts were formed (Pt/Au/Pd for p-FET and Au for n-FET). Another WSe₂ n-FET was achieved by Yu et al. through the low-work-function metal (Ag) contact and tetracyanoquinodimethane (F₄TCNQ) n-doping technique.^[17] However, all researches currently remain at the level to accomplish simultaneously the different types of devices on one kind of TMDs. In particular, the performance of type-converted FET devices has not been clearly investigated in terms of optoelectronic devices.

Here, we report a high performance TMD photodetector with both a fast temporal photoresponse and a high responsivity that are in trade-off. This performance is achieved in a WSe₂-based FET fabricated on a hexagonal boron nitride (h-BN) layer by: i) converting the device type from a p- to n-channel and ii) precisely controlling the doping concentration of the WSe₂ layer through a triphenylphosphine (PPh₃)-based n-doping technique that is newly developed in this work. The phosphorus atoms in the PPh₃ molecules form lone pairs of electrons that enable the donation of electrons to the WSe₂ layer at level of 10¹¹ cm^{−2}. First, we discuss the controllability of PPh₃ n-doping on WSe₂ with Raman spectroscopy and X-ray photoelectron spectroscopy (XPS). The effects of PPh₃ n-doping are then investigated on WSe₂-based FETs with different metal contacts (Ti and Pt) in terms of the performance of electronic (threshold voltage, on-current level, carrier concentration, and field-effect mobility) and optoelectronic (photoresponsivity and temporal photoresponse) devices. Finally, we further improve the photoresponsivity and temporal photoresponse performance of the PPh₃-doped WSe₂ photodetector by inserting the h-BN layer underneath the channel area and consequently suppressing the scattering phenomenon at the WSe₂/SiO₂ interface.

First, we prepared three different PPh₃ concentration solutions (2.5, 5.0, and 7.5 wt%) and WSe₂ samples, which were mechanically exfoliated onto a SiO₂/Si substrate by scotch-tape in order to investigate the effects of PPh₃ doping on the WSe₂ by Raman spectroscopy and XPS analyses. As shown in Figure 1a, two conventional Raman peaks (E_{12g} and A_{1g}) were observed near 250 and 260 cm^{−1} in the undoped WSe₂ sample (gray line), where the peaks indicate the in- and out-of-plane vibrations for bulk WSe₂, respectively.^[18] The positions of the E_{12g} and A_{1g} Raman peaks were slightly blueshifted (dotted color lines) after performing the PPh₃ n-doping process on the WSe₂ films. We then extracted the peak shift values in each Raman peak before and after PPh₃ n-doping, and the values were plotted as a function of the PPh₃ concentrations in Figure 1b. For each

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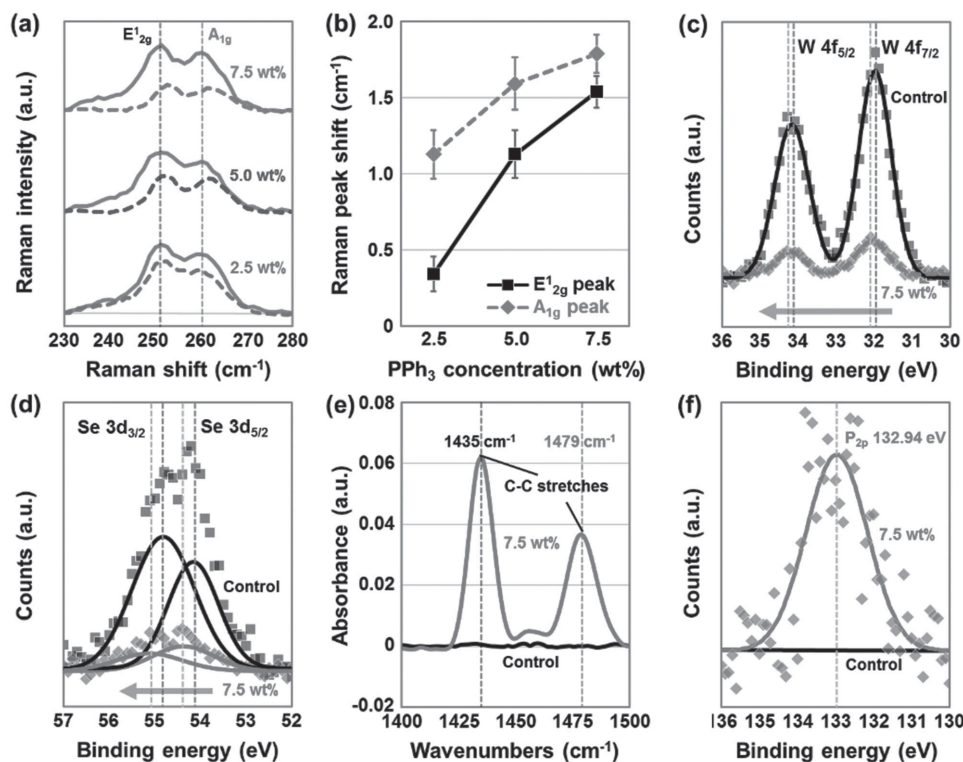


Figure 1. Characterization of the WSe₂ films before/after PPh₃ doping. a) Raman spectra of the control (solid line) and PPh₃-doped (dotted line) WSe₂ at different concentrations (2.5, 5.0, and 7.5 wt%). b) Extracted Raman peak shift values of WSe₂ doped by PPh₃ as a function of the PPh₃ concentration. XPS analysis of WSe₂ before and after 7.5 wt% PPh₃ doping with binding energy peaks of: c) W 4f_{5/2} and 4f_{7/2} and d) Se 3d_{3/2} and 3d_{5/2} electrons. e, f) C–C stretch peaks and P 2p electron binding energy peaks appeared in FTIR (e) and XPS (f) spectra after 7.5 wt% PPh₃ doping.

doping condition, eight different samples were prepared, and the Raman measurements were performed at three different points in each sample. As shown in Figure 1b, blueshifting of the E_{12g} and A_{1g} peaks was observed in the ranges of 0.34–1.54 and 1.13–1.79 cm^{−1}, respectively. The blueshifting (for E_{12g} and A_{1g} peaks) became clearer as the PPh₃ concentration increased. In previous Raman studies on WSe₂, an erratic Raman peak shift was reported after the p-doping process by various materials: blueshift by octadecyltrichlorosilane,^[19] redshift by Au nanoparticles,^[20] and no-shift by NO_x.^[21] Because it was not obvious that Raman analysis could confirm the type of doped WSe₂, we then performed XPS measurements on the undoped/PPh₃-doped WSe₂ films. Figure 1c,d shows the binding energy peaks of W 4f and Se 3d before and after the 7.5 wt% PPh₃ n-doping on the WSe₂ film. After the PPh₃ n-doping, the W 4f and Se 3d peaks were upshifted by 0.08 and 0.3 eV, respectively, indicating that the Fermi level of WSe₂ moved toward the conduction band edge (n-type doping phenomenon).^[16,22] The previous XPS analysis of potassium n-doped WSe₂ reported by Fang et al. also showed an up-shift (≈0.4 eV) of the electron binding energy peaks.^[16] In addition, FT-IR and XPS analyses were used to verify the characteristics of the PPh₃ layers on the WSe₂ flakes. As shown in Figure 1e, two conventional C–C stretch peaks were observed at 1435 cm^{−1} and 1479 cm^{−1}, which indicate phenyl groups (C₆H₅–) in the PPh₃ atomic structure. The existence of phosphorus atoms in the PPh₃ was also confirmed through the XPS spectra of the phosphates (P 2p peak at 132.94 eV) in Figure 1f.

In order to investigate the PPh₃ n-doping effects on the electrical performance of WSe₂ FETs, we then fabricated back-gated WSe₂ FET devices with two different metal contacts (titanium and platinum) and performed current–voltage measurements (I_D – V_G and I_D – V_D) in air. Figure 2a,b shows the schematic illustration, the optical image of the PPh₃-treated WSe₂ device (both the channel length and width are 5 μm), and the detailed description of the n-doping mechanism at the PPh₃/WSe₂ interface. As described in Figure 2b, when the PPh₃ layer was coated on the WSe₂ devices, the electron lone pair of PPh₃ is expected to induce a negative charge transfer phenomenon from PPh₃ to WSe₂, thereby shifting up the Fermi level of the WSe₂. This phenomenon consequently increases an effective hole barrier height at the Pt/WSe₂ junction and decreases the hole injection probability from the source (Pt) to the channel (WSe₂). In contrast, the up-shift of the WSe₂ Fermi level by the PPh₃ n-doping seems to reduce the effective electron barrier height at the Ti–WSe₂ junction, increasing the electron injection probability from Ti to the WSe₂ channel. As shown in Figure 2c–j, we then extracted various electrical parameters (ΔV_{TH} , Δn , μ_{FE} , and I_{on}) and evaluated the PPh₃-doped Ti- and Pt–WSe₂ devices with applied PPh₃ concentrations of 2.5, 5.0, and 7.5 wt%. For the extraction, the field-effect mobility formula $\mu_{FE} = L/(WV_D C_{OX}) \times (\partial I_D / \partial V_G)$ and 2D sheet doping concentration formula n (or p) = $I_D L / q W \mu V_D$ were used, where q is the electron charge, L and W are the length and width of the channel, respectively, and the gate oxide capacitance per unit area (C_{OX}) is $\epsilon_{OX} \times \epsilon_0 / t_{OX}$. μ_{FE} was calculated at $V_{GS} - V_{TH} = 0$ V,

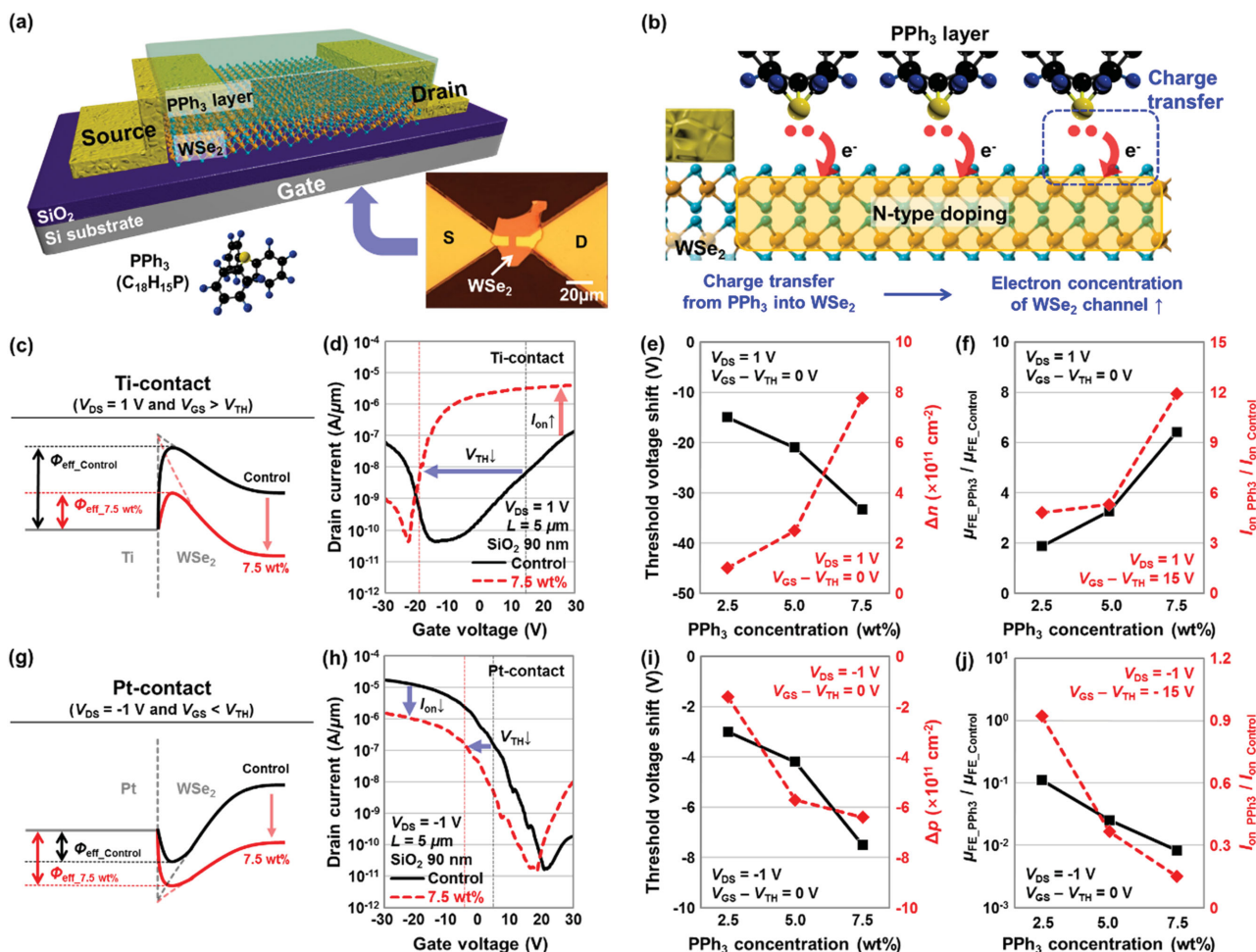


Figure 2. Electrical characterization of PPh₃-doped Ti- and Pt-WSe₂ transistors. a) Schematic diagram and optical image of back-gated (90-nm thick SiO₂ gate dielectric) WSe₂ transistor device doped by PPh₃. b) Descriptive diagram for the n-doping mechanism of PPh₃ at the PPh₃/WSe₂ interface. c, g) Energy-band diagram of Ti-WSe₂ (c), Pt-WSe₂ (g) junctions at on-state before and after 7.5 wt% PPh₃ doping. d, h) I_D - V_G characteristics of control (black solid line) and 7.5 wt% PPh₃-doped (red dotted line) WSe₂ transistors ($L = 5 \mu\text{m}$) with Ti-contact ($V_{\text{DS}} = 1 \text{ V}$) (d) or Pt-contact ($V_{\text{DS}} = -1 \text{ V}$) (h). e, i) The extracted threshold voltage shifts ($\Delta V_{\text{TH}} = V_{\text{TH, PPh}_3} - V_{\text{TH, Control}}$) and carrier concentration differences ($\Delta n = n_{\text{PPh}_3} - n_{\text{Control}}$ and $\Delta p = p_{\text{PPh}_3} - p_{\text{Control}}$) at $V_{\text{GS}} = V_{\text{TH}}$ in the WSe₂ transistors with Ti-contact ($V_{\text{DS}} = 1 \text{ V}$) (e) or Pt-contact ($V_{\text{DS}} = -1 \text{ V}$) (i). f, j) The extracted field-effect mobility ratio (μ_{FE} ratio = $\mu_{\text{FE, PPh}_3} / \mu_{\text{FE, Control}}$) at $V_{\text{GS}} = V_{\text{TH}}$ and on-current ratio (I_{on} ratio = $I_{\text{on, PPh}_3} / I_{\text{on, Control}}$) at $V_{\text{GS}} = V_{\text{TH}} + 15 \text{ V}$ (n-FET) and $V_{\text{GS}} = V_{\text{TH}} - 15 \text{ V}$ (p-FET) in WSe₂ transistors with Ti-contact ($V_{\text{DS}} = 1 \text{ V}$) (f) or Pt-contact ($V_{\text{DS}} = -1 \text{ V}$) (j).

and I_{on} was extracted at $V_{\text{GS}} - V_{\text{TH}} = 15 \text{ V}$ (Ti-contacted device) and $V_{\text{GS}} - V_{\text{TH}} = -15 \text{ V}$ (Pt-contacted device). We note that the contact resistance was not clearly excluded in the μ_{FE} calculation so that the extracted μ_{FE} values were dependent on the contact resistance (also, carrier injection at the metal-TMD junction interface) and slightly underestimated due to the considerable contact resistance.^[23] Figure 2c, g shows the energy-band diagrams of the Ti- and Pt-WSe₂ junctions at on-state before/after the PPh₃ n-doping process by various concentrations of PPh₃ solutions. In the case of the Ti-WSe₂ junctions (Figure 2c), the electron carrier injection from Ti to WSe₂ is expected to improve as the PPh₃ concentration increases due to the reduction in effective electron barrier height. As a result, the extracted V_{TH} value shifted from 14.1 to -19.2 V and the on-current (at $V_{\text{GS}} - V_{\text{TH}} = 15 \text{ V}$) increased from 1.24×10^{-7} to $1.47 \times 10^{-6} \text{ A } \mu\text{m}^{-1}$ after 7.5 wt% PPh₃ doping, as shown in Figure 2d. The corresponding I_D - V_D characteristic curves are

shown in Figure S1 in the Supporting Information, which also indicates that the PPh₃-doped Ti-contacted WSe₂ device was successfully type-converted to the n-channel device. Figure 2e, f shows the electrical parameters (V_{TH} , Δn , μ_{FE} , and I_{on}) as a function of the PPh₃ concentrations, which were extracted from the Ti-contacted devices. When the PPh₃ concentration increased, ΔV_{TH} decreased from -15 to -33.3 V and Δn increased from 1.02×10^{11} to $7.77 \times 10^{11} \text{ cm}^{-2}$. These Δn values were observed in the nondegenerate level, which are very similar to the levels of the previously reported OTS p-doping^[19] ($\approx 2.3 \times 10^{11} \text{ cm}^{-2}$) or Cs₂CO₃ n-doping^[24] ($\approx 3.5 \times 10^{11} \text{ cm}^{-2}$). Additionally, as shown in Figure 2f, μ_{FE} and I_{on} were respectively improved from 1.88 to 6.42 (μ_{FE} ratio) and from 4.81 to 11.9 (I_{on} ratio) with increasing PPh₃ concentration, where the highest μ_{FE} and I_{on} values were $18.9 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ and $4.0 \times 10^{-6} \text{ A } \mu\text{m}^{-1}$, respectively, in the 7.5 wt% PPh₃-doped Ti-WSe₂ device. This enhancement in μ_{FE} and I_{on} seems attributable to the increase in the electron carrier

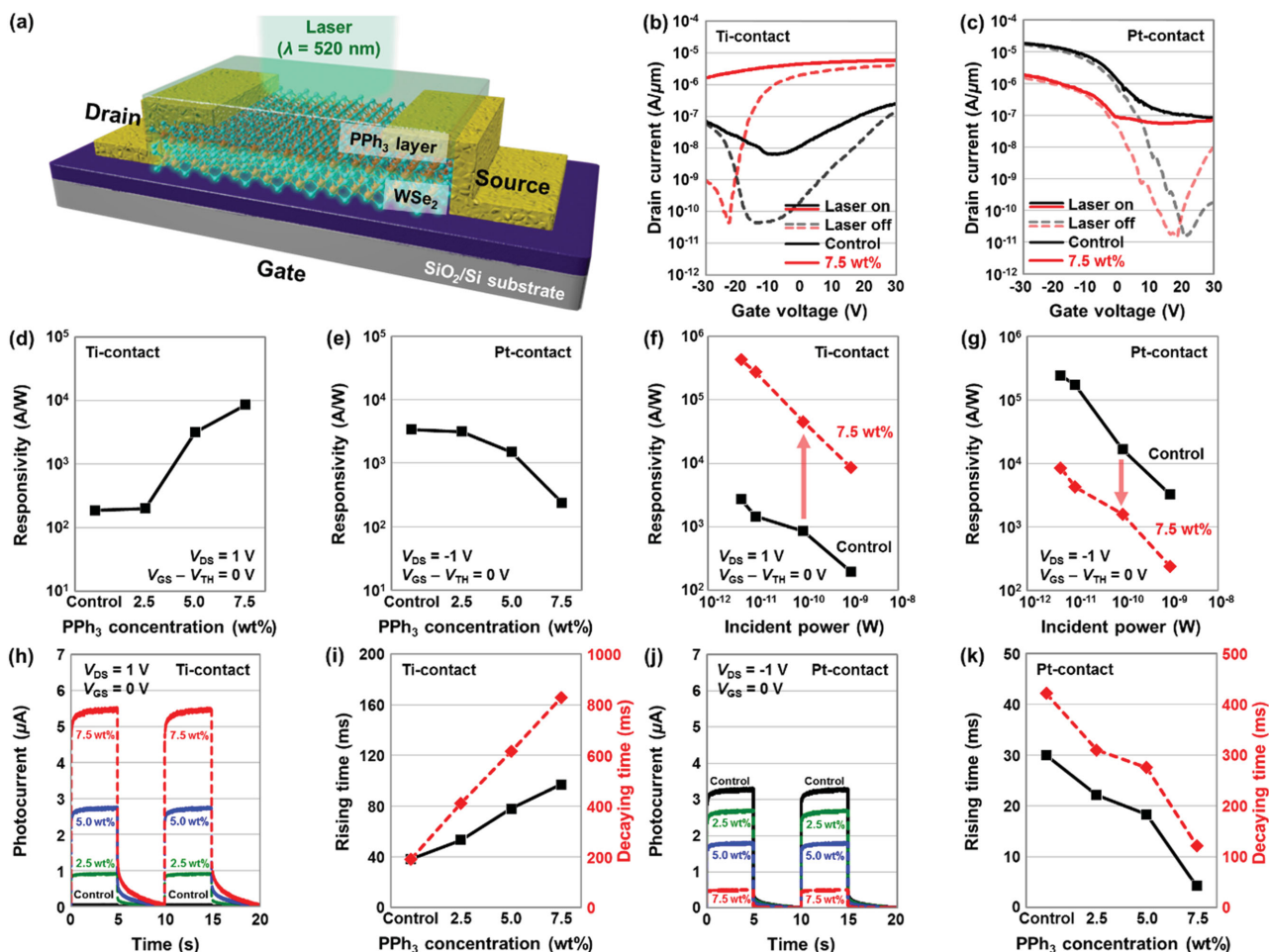


Figure 3. Optoelectronic characterization of PPh₃-doped Ti- and Pt-WSe₂ photodetectors. a) Schematic of the PPh₃-doped WSe₂ photodetector under laser illumination ($P_{\text{light}} = 4 \text{ mW cm}^{-2}$ and $\lambda = 520 \text{ nm}$). b, c) I_D - V_G characteristics of the control (black) and PPh₃-doped (red) photodetectors under both dark (dotted line) and illuminated (solid line) conditions with Ti-contact ($V_{DS} = 1 \text{ V}$) (b) or Pt-contact ($V_{DS} = -1 \text{ V}$) (c). d, e) Photoresponsivity as a function of the PPh₃ doping concentration, which was extracted at $V_{GS} = V_{TH}$ in the photodetectors with Ti-contact ($V_{DS} = 1 \text{ V}$) (d) or Pt-contact ($V_{DS} = -1 \text{ V}$) (e). f, g) Photoresponsivity as a function of incident laser power (5, 10, 100, and 1000 pW) was extracted at $V_{GS} = V_{TH}$ before and after 7.5 wt% PPh₃-doping in the WSe₂ photodetectors with Ti-contact ($V_{DS} = 1 \text{ V}$) (f) or Pt-contact ($V_{DS} = -1 \text{ V}$) (g). h-k) The temporal photoresponse characteristics and rising/decaying times of the photodetectors were obtained at $V_{GS} = 0 \text{ V}$ with Ti-contact ($V_{DS} = 1 \text{ V}$) (h,i) or Pt-contact ($V_{DS} = -1 \text{ V}$) (j,k), respectively.

injection probability as the PPh₃ concentration increases. Compared to the Ti-contacted WSe₂ devices, the PPh₃ n-doping oppositely influenced the electrical characteristics in the Pt-contacted WSe₂ devices. As shown in Figure 2g, the hole carrier injection from Pt to WSe₂ seems to deteriorate as the PPh₃ concentration increases, due to the increase in the effective hole barrier height at the Pt-WSe₂ junction. Consequently, in the 7.5 wt% PPh₃-doped Pt-contacted devices, V_{TH} shifted from 4.5 to -3 V and the on-current (at $V_{GS} - V_{TH} = -15 \text{ V}$) decreased from 5.84×10^{-6} to $8.78 \times 10^{-7} \text{ A } \mu\text{m}^{-1}$ (Figure 3h). We also extracted and compared the electrical parameters with varying PPh₃ concentration, as shown in Figure 2i,j. Although the extracted ΔV_{TH} and Δp values were similarly changed as a function of the PPh₃ concentration like the case of the Ti-contacted devices showing n-doping behaviors (ΔV_{TH} : $-3.0 \rightarrow -7.5 \text{ V}$ and Δp : $-1.6 \times 10^{11} \rightarrow -6.4 \times 10^{11} \text{ cm}^{-2}$ in Figure 2e), the μ_{FE} and I_{on} ratio values were respectively reduced from 1.11×10^{-1} to $8.26 \times 10^{-3} \text{ cm}^{-2}$ and from 9.21×10^{-1} to $1.50 \times 10^{-1} \text{ A } \mu\text{m}^{-1}$ as

the PPh₃ concentration increased. In contrast to the performance enhancement in the Ti-contacted devices, the electrical performance of the Pt-contacted devices degraded with increasing PPh₃ concentration. This degradation can also be explained by the decrease in the hole carrier injection probability, owing to the increase in effective hole barrier height at the Pt-WSe₂ junction with increasing PPh₃ concentration. We also confirmed the controllability of PPh₃ n-doping in both Ti- and Pt-contacted WSe₂ devices by adjusting the PPh₃ concentration, which is consistent with the Raman analysis (Figure 1a,b). In addition, we investigated the stability of the PPh₃ doping effects on Ti- and Pt-contacted WSe₂ devices with various concentrations (2.5, 5.0, and 7.5 wt%, Figure S2, Supporting Information). After 24 h in air, the threshold voltage shifts of 2.5 wt% and 5.0 wt% PPh₃-doped samples decreased ≈ 10.1 –53.1% in Ti-WSe₂ and 22.6–33.9% in Pt-WSe₂, respectively. In particular, after 7.5 wt% PPh₃ doping, very small threshold voltage shifts were observed after 192 h in air (3.83% and 2.73% in the Ti- and

Pt-WSe₂ samples, respectively), indicating that the 7.5 wt% PPh₃-doped FET device was remarkably stable compared to the devices doped by other techniques, such as OTS^[19] (ΔV_{TH} : 22.3% after 120 h).

Next, we investigated the PPh₃ n-doping effects on the optoelectronic performance of WSe₂ photodetectors by performing electrical measurements in air under dark and laser illuminated conditions ($\lambda = 520$ nm). Figure 3a shows the schematic illustration of the PPh₃-doped WSe₂ photodetector with different contact electrodes (Ti and Pt). When the PPh₃ layer was applied to the Ti-contacted devices, the contact resistance decreased because of the reduction in the effective electron-injection barrier height. This eventually appears to improve the photocurrent by increasing the electric field at the WSe₂ region, as shown in Figure 3b. After the Ti-contacted WSe₂ device was doped with 7.5 wt% PPh₃, the photocurrent increased from 5.87×10^{-8} to 2.59×10^{-6} A μm^{-1} , which was extracted at $V_{GS} = V_{TH}$. In contrast, the photocurrent of the Pt-contacted WSe₂ photodetector decreased after the PPh₃ n-doping (Figure 3c) due to the reduced photocarrier collection at the WSe₂ region by the increase in the Pt-WSe₂ contact resistance. In order to investigate the detailed PPh₃ doping effects on the WSe₂ photodetectors, the photoresponsivity ($R = I_{\text{Photo}}/P_{\text{Light}}$) was then extracted before and after the PPh₃ n-doping with different concentrations (2.5, 5.0, and 7.5 wt%). Here, the generated photocurrent (I_{Photo}) is $I_{\text{Laser_on}} - I_{\text{Laser_off}}$, and P_{Light} is the total incident laser power. The photoresponsivity extractions were performed when V_{GS} was equal to V_{TH} (Figure S3, Supporting Information). In the case of the control devices (undoped WSe₂ photodetector), the photoresponsivity values were $\approx 1.86 \times 10^2$ A W⁻¹ (Ti-contacted devices) and 3.39×10^3 A W⁻¹ (Pt-contacted devices). Because of the opposite-behavior in the contact resistances at source-WSe₂ junctions by PPh₃ n-doping, specifically, a decrease (Ti-contacted devices) and an increase (Pt-contacted device) in contact resistance (above Figure 2), the photocarrier collection is expected to be suppressed and enhanced in the Ti- and Pt-contacted devices, respectively. Higher contact resistance at the metal-TMD junction interface seems to reduce the effective electric field in the TMD region. When the PPh₃ layer doped the Ti-contacted photodetector, an increasing trend in the photoresponsivity was observed as a function of the PPh₃ concentration from 1.86×10^2 (control) to 8.63×10^3 A W⁻¹ (7.5 wt%). Alternatively, the photoresponsivity of the doped Pt-contacted devices decreased as the PPh₃ concentration increased ($3.39 \times 10^3 \rightarrow 2.37 \times 10^2$ A W⁻¹ at 7.5 wt%).

Figure 3f–k shows the additional analyses of the PPh₃-doped WSe₂ photodetectors, which are related to the total incident power (f–g) and temporal photoresponse (h–k). According to previous report,^[25] the photoresponsivity of the photodetectors increased as an inverse exponential function of the incident power of the laser because the scattering effect among the photogenerated carriers was suppressed under low-power laser illumination. The photocurrents were measured after exposing the lasers with different incident powers (5, 10, 100, and 1000 pW) and then the photoresponsivity values were extracted as a function of the incident power. In the case of Ti-contacted devices (Figure 3f), the photoresponsivity increased by a factor of 13.81 (control device) and 49.91 (7.5 wt% PPh₃-doped device) as the incident power decreased, where the maximum photoresponsivity was

4.31×10^5 A W⁻¹ (under 5 pW in the 7.5 wt% PPh₃-doped device). A similar improvement in photoresponsivity by reducing the incident power of the laser was also observed in the Ti-contacted devices (Figure 3g), but the photoresponsivity values of the PPh₃-doped devices were much less than those of the Ti-contacted devices due to the negative effect of PPh₃ doping on the Pt-WSe₂ contact ($2.43 \times 10^5 \rightarrow 8.52 \times 10^3$ A W⁻¹, under 5 pW laser). In addition, the temporal photoresponse in the PPh₃-doped WSe₂ photodetectors was measured with laser-switching cycles of 5 s. The extraction method for the rising and decaying times is explained in detail in Figure S4 in the Supporting Information. Figure 3h,j presents the temporal photoresponse curves of the Ti- and Pt-contacted photodetectors doped by various concentrations of PPh₃. In the case of Ti-contacted devices, the photoresponse times increased (τ_r : 38.2 $\rightarrow$ 97 ms and τ_d : 191 $\rightarrow$ 827 ms) as the PPh₃ concentration increased. Alternatively, an improvement in the rising and decaying times was observed in the PPh₃-doped and Pt-contacted devices (τ_r : 30 $\rightarrow$ 4.2 ms and τ_d : 422 $\rightarrow$ 122 ms). This conflicting photoresponse can be explained by the different variations in the depletion widths of the Ti- and Pt-WSe₂ junctions by PPh₃-doping. The depletion width of the Ti-WSe₂ junction is expected to narrow as the PPh₃ concentration increases, as opposed to the expanded depletion width at the Pt-WSe₂ junction. Because the narrowed depletion width makes it difficult to collect the photogenerated carriers in the WSe₂ region, the photoresponse (rising and decaying times) of the PPh₃-doped and Ti-contacted photodetectors becomes slower than the undoped devices, which is opposite to the Pt-WSe₂ devices which show a faster photoresponse after the PPh₃ n-doping.

By converting the WSe₂ device type (p- to n-channel) through PPh₃ n-doping and Ti contact formation, we could achieve a high photoresponsivity that is similar to (or slightly greater than) that of the Pt-WSe₂ photodetector (Ti-WSe₂: 4.31×10^5 A W⁻¹ and Pt-WSe₂: 2.43×10^5 A W⁻¹), but the temporal photoresponse times (τ_r : 97 ms and τ_d : 827 ms) were 2–3 times degraded compared to the Pt-WSe₂ device (τ_r : 30 ms and τ_d : 422 ms). Therefore, in order to recover the degraded temporal photoresponse time of the photodetector and eventually make the n-channel device performance comparable to the p-channel WSe₂ photodetector, an *h*-BN layer was employed to the Ti-contacted n-channel WSe₂ photodetector. Because the *h*-BN layer provides a charge-free environment for the channel region, it can enhance the carrier mobility of the TMD FET devices by inserting at TMD/SiO₂ interface^[26,27] or encapsulating the TMD surfaces.^[28,29] Figure 4a shows the schematic diagrams and the optical microscopy images of the PPh₃-doped WSe₂ device with *h*-BN between the WSe₂ and SiO₂ layers. Here, we used an inverse transfer technique for fabricating the WSe₂/*h*-BN heterostructure-based device (Figure S5, Supporting Information) to achieve a clean WSe₂/*h*-BN interface. By inserting *h*-BN between the WSe₂ channel and the SiO₂ gate dielectric, as shown in Figure 4b, the photoresponsivity of the PPh₃-doped Ti-contacted WSe₂ photodetector was increased by a factor of 1.9 ($6.67 \times 10^5 \rightarrow 1.27 \times 10^6$ A W⁻¹ under 5 pW laser exposure). Here, we compared the maximum photoresponsivity values obtained in the same range of gate voltage (between -30 V and 30 V). For reference, this increasing trend in photoresponsivity after *h*-BN insertion will be still valid even after

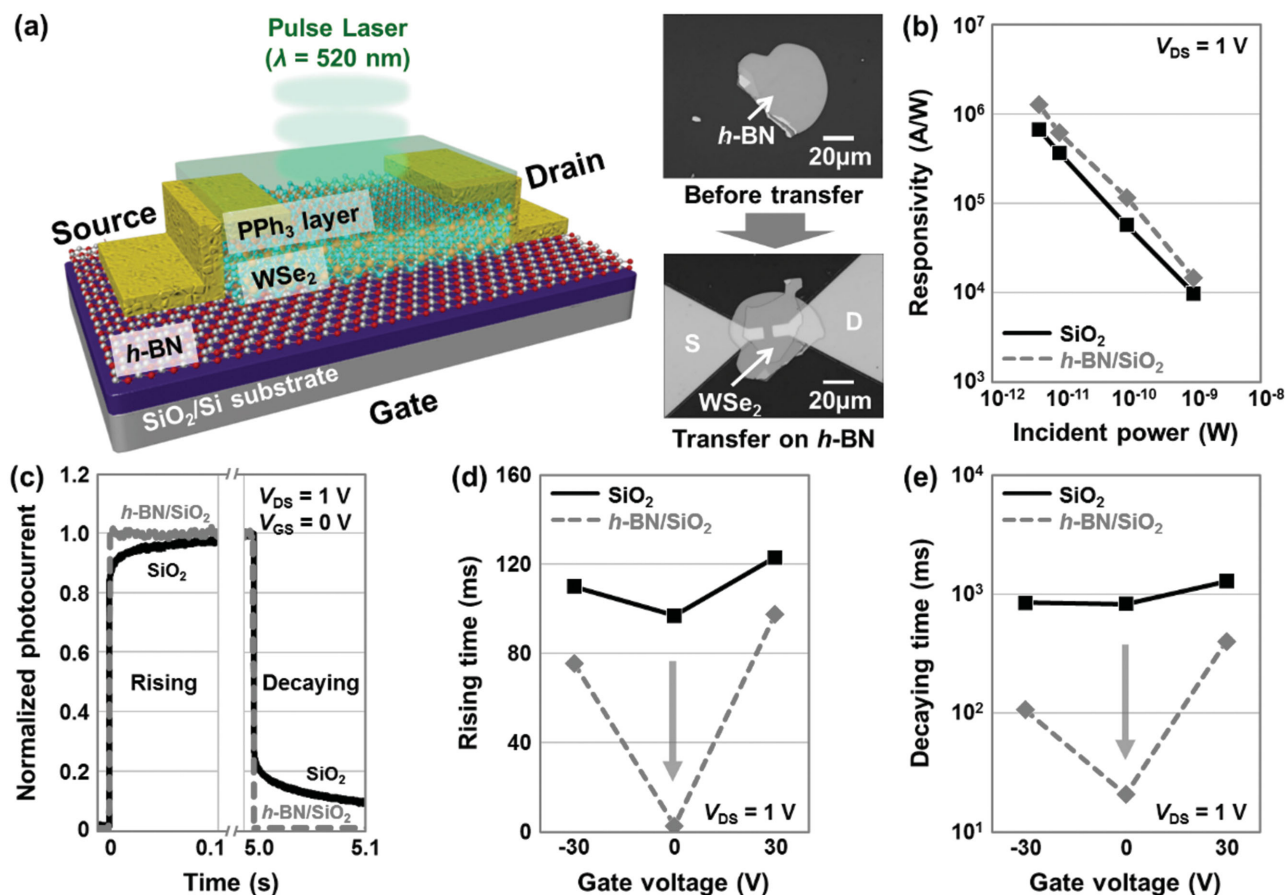


Figure 4. Characterization of PPh₃-doped Ti-WSe₂/h-BN photodetectors. a) Schematic diagram and optical images (before and after transfer) of h-BN inserted into the Ti-contacted WSe₂ photodetector that was doped by 7.5 wt% PPh₃. b) Extracted photoresponsivity as a function of the incident laser power (5, 10, 100, and 1000 pW) before and after transferring to the h-BN/SiO₂ substrate. c) Normalized temporal photoresponse curves at the rising and decaying edges. d,e) Rising (d) and decaying (e) times of the photodetector as a function of the gate voltage ($V_G = -30, 0$, and 30 V), where the drain voltage (V_{DS}) was 1 V.

equalizing the gate capacitances because the smaller vertical electric field in the h-BN-inserted photodetectors (compared to the control devices) is expected to induce weaker photoresponsivity. This improvement can be attributed to the increased carrier mobility (from $18.9 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ before the insertion of h-BN to $25.6 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ after the insertion) and the subsequently enhanced carrier collection, which were obtained by separating the WSe₂ channel region from the charged surface of SiO₂.^[26] As the incident laser power was reduced from 1000 to 5 pW, the photoresponsivity increased by approximately two orders of magnitude as a function of the inverse of the power, due to the suppressed scattering probability of the photo-generated carriers. In addition, the temporal responses of the PPh₃-doped Ti-contacted WSe₂ photodetectors with and without an h-BN layer were investigated. The temporal photoresponse curves in Figure 4c ($V_{DS} = 1$ V and $V_{GS} = 0$ V) clearly show that the rising and decaying time were improved after inserting the h-BN layer. As shown in the extracted rising (Figure 4d) and decaying (Figure 4e) time values, enhanced temporal responses were observed in all ranges of the applied gate voltage (between -30 V and 30 V). This enhancement may also be caused by the improved carrier mobility via insertion of the

h-BN layer because high mobility will boost the transporting speed of the photocarriers. The shortest temporal response time was observed at $V_G = 0$ V, and the rising and decaying times were respectively enhanced from 97 to 2.8 ms and 827 to 20.8 ms by applying the h-BN layer, where the improved values were much shorter than those of Pt-contacted WSe₂ control device (τ_r : 30 ms and τ_d : 422 ms). By applying the h-BN layer to the PPh₃-doped Ti-contacted WSe₂ photodetector, we successfully recovered the temporal photoresponse time, where the obtained photoresponse time in milliseconds was similar to the value (1.8–2.0 ms) of the photodetector fabricated on MoS₂ by using a ferroelectric material, P(VDF-TrFE), as a gate insulator.^[30] For reference, when a 2D material-based photodetector was fabricated in vertical hybrid structure consisting of GaTe_xSe_{1-x} and Si, its rising and decaying times were relatively lower as much as 130 and 70 μs, respectively.^[31] In addition, we confirmed that the temporal photoresponse time was affected by an applied gate voltage bias in all photodetectors. According to Figure 4d,e and Figure S6 (Supporting Information), the rising and decaying times were degraded at $V_G = \pm 30$ V compared to the zero gate bias ($V_G = 0$ V). This phenomenon seems to be caused by the electron/hole carrier accumulation that

is formed when a positive or negative gate voltage is applied, which consequently occurs as a carrier scattering effect in the WSe₂ channel region and thereby reduces the mobility of the photogenerated carriers.

In conclusion, we demonstrated a WSe₂-based photodetector with high responsivity and fast switching speed through: i) FET-type conversion (n- to p-channel) via a PPh₃ n-doping technique, and ii) *h*-BN insertion underneath the WSe₂ channel. Especially, the obtained maximum photoresponsivity was a record value up to present, compared to other single TMD material-based photodetectors. The n-doping originated from the electron transfer phenomenon from the lone pair of phosphorus atoms in PPh₃. The achieved n-doping concentration (Δn and Δp) on the WSe₂ films was in a nondegenerate regime, presenting values between $1.02 \times 10^{11} \text{ cm}^{-2}$ and $7.77 \times 10^{11} \text{ cm}^{-2}$ (Δn in Ti-contacted WSe₂ device) and between $-1.61 \times 10^{11} \text{ cm}^{-2}$ and $-6.4 \times 10^{11} \text{ cm}^{-2}$ (Δp in Pt-contacted WSe₂ device). We also investigated the PPh₃ doping effects in terms of the performance of WSe₂-based electronic and optoelectronic devices (V_{TH} , μ_{FE} , I_{ON} , photoresponsivity, and temporal photoresponse) by adjusting the doping concentrations. By forming Ti contacts and performing a 7.5 wt% PPh₃ n-doping process, we completely converted the WSe₂ FET from a p- to n-channel, and also achieved a comparable performance to that of the Pt-contacted WSe₂ p-FET device. In the case of the photodetector, the photoresponsivity increased ($2.70 \times 10^3 \rightarrow 4.31 \times 10^5 \text{ A W}^{-1}$ under 5 pW laser exposure), but the temporal photoresponse deteriorated (τ_r : 38.2 $\rightarrow$ 97 ms and τ_d : 191 $\rightarrow$ 827 ms) after 7.5 wt% PPh₃ n-doping and Ti contact formation. The poor temporal photoresponse was then recovered by applying the *h*-BN layer at the WSe₂/SiO₂ interface and eventually suppressing the carrier scattering at the channel region. As a result, the maximum photoresponsivity was enhanced from 6.67×10^5 to $1.27 \times 10^6 \text{ A W}^{-1}$ under 5 pW laser exposure, and the rising and decaying times were also recovered from 97 to 2.8 ms and from 827 to 20.8 ms, respectively, which are comparable to or better than the performance of the Pt-contacted WSe₂ photodetector. Furthermore, we confirmed that the PPh₃ doping was very stable in air with only 3.83% (Ti contact) and 2.73% (Pt contact) variations in V_{TH} observed for 192 h in the case of the 7.5 wt% PPh₃-doped WSe₂ devices.

Experimental Section

PPh₃ n-Doping of WSe₂: For preparing the PPh₃ solutions, different amounts of triphenylphosphine (0.25, 0.50, and 0.75 g) were respectively dissolved in various amounts of toluene (9.75, 9.50, and 9.25 g) to produce doping solutions with various concentrations of PPh₃, respectively, named as PPh₃ 2.5, 5.0, and 7.5 wt%. For doping WSe₂, the PPh₃ solution was spin-coated on the prepared WSe₂ samples at 1000 rpm for 30 s (5 s acceleration). The samples were then annealed at 150 °C for 10 min to remove the remaining solvent.

Raman, XPS, and FT-IR Analyses of PPh₃-Doped WSe₂: The characteristics of the WSe₂ (which was mechanically exfoliated and transferred to the SiO₂/Si substrate) were investigated by Raman spectroscopy (Alpha300 M+, WITec), XPS (ESCA200, VG Microtech Inc.), and FTIR (IFS-66/S, Bruker) before and after PPh₃ n-doping. Raman spectroscopy with an excitation wavelength of 532 nm was used with a laser beam size of 0.7–0.9 μm and an instrumental spectral resolution less than 0.9 cm^{-1} . The integration time was 5 s and a spectrometer

with 1800 grooves mm^{-1} was employed for the test. A Mg K α twin-anode source was used for the XPS measurement, where the X-ray incident angle was 0°. The FTIR spectral range was between 4000 and 20 cm^{-1} , the scan rate was 110 scans s^{-1} , and the resolution was greater than 0.1 cm^{-1} .

Fabrication of PPh₃-Doped WSe₂ Electronic and Optoelectronic Devices: The WSe₂ flake was mechanically exfoliated and transferred to the 90 nm thick, SiO₂ oxidized, heavily boron-doped Si substrate via adhesive tape (224SPV, Nitto). The remaining adhesive tape residue was removed by immersing the samples in an acetone bath for 1 h. For the fabrication of the back-gated WSe₂ devices (transistors and photodetectors), source/drain electrode regions (channel length and width were 5 μm) were patterned on the WSe₂/SiO₂/Si samples by an optical lithography process, followed by 10 nm thick Ti (for *n*-type FET) or Pt (for *p*-type FET) and 50-nm thick Au deposition processes via an e-beam evaporator. Additionally, the fabricated WSe₂ devices were transferred to the *h*-BN/SiO₂/Si substrate by an inverse transfer method using poly(methyl methacrylate) (PMMA) and poly(vinyl alcohol) (PVA) supporting layers (Figure S5, Supporting Information).

Electrical Characterization of the PPh₃-Doped WSe₂ Electronic Device: The fabricated devices were electrically analyzed through electrical measurements ($I_{\text{D}}-V_{\text{G}}$ and $I_{\text{D}}-V_{\text{D}}$) by an HP 4155A semiconductor parameter analyzer. The threshold voltage (V_{TH}), carrier concentration (*n* or *p*), and field-effect mobility (μ_{FE}) were calculated from the measured data. Here, all drain currents (I_{D}) were normalized by the channel width (*W*). The carrier concentration and field-effect mobility were respectively extracted using *n* (or *p*) = $I_{\text{D}}L/qW\mu_{\text{FE}}V_{\text{D}}$ and $\mu_{\text{FE}} = L/(WV_{\text{D}}C_{\text{OX}}) \times (\partial I_{\text{D}}/\partial V_{\text{G}})$, where *L* and *W* are the length and width of the channel, respectively, *q* is the electron charge, and C_{OX} is $\epsilon_{\text{OX}} \times \epsilon_0/t_{\text{OX}}$, which is the gate oxide capacitance per unit area.

Optical Characterization of the PPh₃-Doped WSe₂ Optoelectronic Device: The PPh₃-doped WSe₂ photodetector devices were investigated by performing electrical measurements ($I_{\text{D}}-V_{\text{G}}$) under both dark and illuminated conditions (Figure S7, Supporting Information). The light source was a dot laser with a wavelength of 520 nm and an optical power of 4.0 mW cm^{-2} . The photoresponsivity (*R*) was extracted from $R = I_{\text{photo}}/P_{\text{light}}$ where I_{photo} is the generated photocurrent and P_{light} is the total incident optical power, and the incident power of the laser was adjusted as 5, 10, 100, and 1000 pW. The photoswitching characteristics of the PPh₃-doped WSe₂ photodetector were investigated under the 520 nm laser illumination, which turned on and off with a cycle of 5 s. The photoresponse times (rising and decaying times) were then extracted between 10% and 90% of the measured maximum photocurrent (I_{max}) data (Figure S4, Supporting Information).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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